

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Database Management Systems
COURSE CODE: CSA0509

COURSE OUTCOME COVERED

CO3: *Analyze relational databases using functional dependencies, normalization (1NF to 5NF), and key constraints to identify redundancy and ensure data integrity. (BL2, BL3)*

Activity Tool : Experiments (High)
Assessment Tool Weightage: 40%

QUESTIONS			Total Marks: 50
Q.No	Question	Bloom's Level	Marks
1	For the relation STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.	BL2 – Understand	5
2	Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.	BL3 – Apply	5
3	Given relation R(A,B,C,D,E) with FDs: $A \rightarrow BC$, $BC \rightarrow D$, $D \rightarrow E$. Analyze and decompose it to 2NF eliminating partial dependencies.	BL3 – Apply	5
4	Apply 3NF decomposition on the relation EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) with transitive dependencies.	BL3 – Apply	5
5	Verify whether R(A,B,C,D) with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ is in BCNF. If not, decompose it into BCNF.	BL3 – Apply	5
6	Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.	BL3 – Apply	5
7	Identify the multi-valued dependencies in TEACHER(TID, Subject, Hobby) and decompose it into 4NF.	BL3 – Apply	5

8	Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).	BL3 – Apply	5
9	Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.	BL2 – Understand	5
10	Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.	BL3 – Apply	5

Code of Conduct:

*I Lokesh S (192511037) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

Q1. For the relation STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade), identify all functional dependencies and determine the candidate keys.

Answer:

Given relation:

STUDENT_COURSE(SID, SName, CID, CName, Instructor, Grade)

Functional Dependencies

SID -> SName

CID -> CName

CID -> Instructor

(SID, CID) -> Grade

Explanation:

- SID -> SName: Student ID determines the student's name.
- CID -> CName: Course ID determines the course name.
- CID -> Instructor: Course ID determines the instructor.
- (SID, CID) -> Grade: Grade depends on both the student and the course.
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Finding Candidate Key

Calculate the closure of (SID, CID):

$(SID, CID)^+ = \{SID, CID\}$

Using SID -> SName:

$(SID, CID)^+ = \{SID, CID, SName\}$

Using CID -> CName, Instructor:

$(SID, CID)^+ = \{SID, CID, SName, CName, Instructor\}$

Using (SID, CID) -> Grade:

$(SID, CID)^+ = \{SID, CID, SName, CName, Instructor, Grade\}$

The closure contains all attributes.

Therefore:

Candidate Key = (SID, CID)

Final Answer

Functional Dependencies:

SID -> SName

CID -> CName

CID -> Instructor

(SID, CID) -> Grade

Candidate Key:

(SID, CID)

Q2. Demonstrate the process of converting an unnormalized relation into 1NF with step-by-step justification and example.

Answer:

Consider the unnormalized relation:

STUDENT(SID, SName, Phone, Courses)

Example:

SID	SName	Phone	Courses
S101	Arun	98765	DBMS, OS
S102	Priya	87654	AI, CN

The Courses attribute contains multiple values. This violates 1NF because each cell must contain only one atomic value.

Step 1: Identify the Repeating Group

The repeating attribute is:

Courses

For example:

Courses = DBMS, OS

Step 2: Separate the Values

Each course is placed in a separate row.

SID	SName	Phone	Course
S101	Arun	98765	DBMS

SID	SName	Phone	Course
S101	Arun	98765	OS
S102	Priya	87654	AI
S102	Priya	87654	CN

Step 3: Identify the Primary Key

Since one student can have multiple courses, SID alone is not sufficient.

Therefore:

Primary Key = (SID, Course)

Final 1NF Relation

STUDENT_COURSE(SID, SName, Phone, Course)

The relation is now in 1NF because:

1. Every cell contains a single value.
2. There are no repeating groups.
3. All attributes contain atomic values.
4. Each row can be uniquely identified.

Therefore:

UNF -> 1NF

Q3. Given relation R(A,B,C,D,E) with FDs: A->BC, BC->D, D->E. Analyze and decompose it to 2NF eliminating partial dependencies.

Answer:

Given:

R(A, B, C, D, E)

Functional dependencies:

A -> BC

BC -> D

D -> E

Step 1: Find Candidate Key

Calculate A^+ .

Initially:

$A^+ = \{A\}$

Using:

$A \rightarrow BC$

we get:

$A^+ = \{A, B, C\}$

Using:

$BC \rightarrow D$

we get:

$A^+ = \{A, B, C, D\}$

Using:

$D \rightarrow E$

we get:

$A^+ = \{A, B, C, D, E\}$

Therefore:

$A^+ = \{A, B, C, D, E\}$

So:

Candidate Key = A

Step 2: Check Partial Dependency

A partial dependency occurs when a non-key attribute depends on only part of a composite key.

Here:

Candidate Key = A

The key contains only one attribute.

Therefore, partial dependency is not possible.

Hence, the relation is already in 2NF.

$R(A, B, C, D, E)$ is already in 2NF.

The dependencies $BC \rightarrow D$ and $D \rightarrow E$ indicate further normalization issues, which can be addressed when converting to 3NF.

Final Answer

Candidate Key = A

Partial Dependency = None

Therefore, R is already in 2NF.

Q4. Apply 3NF decomposition on the relation EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID) with transitive dependencies.

Answer:

Given:

EMPLOYEE(EmpID, EmpName, DeptID, DeptName, ManagerID)

Functional dependencies:

EmpID → EmpName, DeptID

DeptID → DeptName, ManagerID

Step 1: Identify Candidate Key

EmpID uniquely identifies an employee.

Candidate Key = EmpID

Step 2: Identify Transitive Dependency

We have:

EmpID → DeptID

DeptID → DeptName, ManagerID

Therefore:

EmpID → DeptName, ManagerID

This is a transitive dependency:

EmpID → DeptID → DeptName, ManagerID

This violates 3NF.

Step 3: Decompose the Relation

Create the employee table:

EMPLOYEE(EmpID, EmpName, DeptID)

Create the department table:

DEPARTMENT(DepID, DeptName, ManagerID)

Final 3NF Decomposition

EMPLOYEE(EmpID, EmpName, DeptID)

DEPARTMENT(DeptID, DeptName, ManagerID)

DeptID in EMPLOYEE is a foreign key referencing DEPARTMENT.

Benefits

- Removes transitive dependency.
- Reduces redundancy.
- Prevents update anomalies.
- Improves data integrity.

Q5. Verify whether $R(A,B,C,D)$ with FDs: $AB \rightarrow C$, $C \rightarrow D$, $D \rightarrow A$ is in BCNF. If not, decompose it into BCNF.

Answer:

Given:

$R(A, B, C, D)$

Functional dependencies:

$AB \rightarrow C$

$C \rightarrow D$

$D \rightarrow A$

Step 1: Find Candidate Keys

AB Closure

$(AB)^+ = \{A, B\}$

Using:

$AB \rightarrow C$

$(AB)^+ = \{A, B, C\}$

Using:

$C \rightarrow D$

$(AB)^+ = \{A, B, C, D\}$

Therefore:

AB is a candidate key.

BC Closure

$(BC)^+ = \{B, C\}$

Using:

$C \rightarrow D$

$(BC)^+ = \{B, C, D\}$

Using:

$D \rightarrow A$

$(BC)^+ = \{A, B, C, D\}$

Therefore:

BC is a candidate key.

BD Closure

$(BD)^+ = \{B, D\}$

Using:

$D \rightarrow A$

$(BD)^+ = \{A, B, D\}$

Using:

$AB \rightarrow C$

$(BD)^+ = \{A, B, C, D\}$

Therefore:

BD is a candidate key.

Candidate Keys

AB

BC

BD

Step 2: Check BCNF

For BCNF, for every non-trivial FD:

$X \rightarrow Y$

X must be a superkey.

Consider:

$C \rightarrow D$

C is not a superkey.

Therefore:

R is NOT in BCNF.

Step 3: Decompose

Use the violating dependency:

$C \rightarrow D$

Create:

$R_1(C, D)$

Remaining relation:

$R_2(A, B, C)$

Final BCNF Decomposition

$R_1(C, D)$

$R_2(A, B, C)$

Q6. Experimentally verify lossless-join decomposition for a given relation using the tabular (Chase) algorithm.

Answer:

Consider:

$R(A, B, C)$

Decomposition:

$R_1(A, B)$

$R_2(B, C)$

Functional dependency:

$B \rightarrow C$

Step 1: Create Chase Table

Relation	A	B	C
R1	a	b	c1
R2	a1	b	c

The common attribute is B.

Step 2: Apply the Functional Dependency

Given:

$B \rightarrow C$

Both rows have the same B value.

Therefore, the C values become equal.

Relation	A	B	C
R1	a	b	c
R2	a1	b	c

Step 3: Verify Lossless Join

If the Chase process produces a row containing all the required distinguished values, the decomposition is lossless.

Therefore:

The decomposition is LOSSLESS.

Conclusion

$R(A, B, C)$

|

+---- $R1(A, B)$

|

+---- $R2(B, C)$

Result: Lossless-Join Decomposition

Q7. Identify the multi-valued dependencies in $TEACHER(TID, Subject, Hobby)$ and decompose it into 4NF.

Answer:

Given:

$TEACHER(TID, Subject, Hobby)$

Suppose a teacher can teach multiple subjects and have multiple hobbies independently.

Example:

TID	Subject	Hobby
T01	DBMS	Cricket
T01	DBMS	Music
T01	AI	Cricket
T01	AI	Music

Step 1: Identify Multivalued Dependencies

A teacher can have multiple subjects independently.

Therefore:

TID $\twoheadrightarrow$ Subject

Similarly:

TID $\twoheadrightarrow$ Hobby

Here $\twoheadrightarrow$ represents a multivalued dependency.

Step 2: Identify 4NF Violation

The relation contains independent multivalued facts about a teacher.

Therefore:

TID $\twoheadrightarrow$ Subject

TID $\twoheadrightarrow$ Hobby

causes a 4NF violation.

Step 3: Decompose

Create:

TEACHER_SUBJECT(TID, Subject)

and:

TEACHER_HOBBY(TID, Hobby)

Final 4NF Decomposition

TEACHER_SUBJECT(TID, Subject)

TEACHER_HOBBY(TID, Hobby)

Result

The decomposition removes unnecessary combinations and reduces redundancy.

Q8. Demonstrate how join dependencies lead to redundancy. Decompose a given relation to 5NF (Project-Join Normal Form).

Answer:

Consider:

SUPPLY(Supplier, Part, Project)

Suppose suppliers, parts and projects have independent relationships.

Example:

Supplier	Part	Project
S1	P1	J1
S1	P1	J2
S1	P2	J1
S1	P2	J2

The same relationships are repeatedly represented, causing redundancy.

Step 1: Decompose the Relation

Create:

SUPPLIER_PART(Supplier, Part)

SUPPLIER_PROJECT(Supplier, Project)

PART_PROJECT(Part, Project)

Step 2: Join Dependency

The original relation can be reconstructed by joining the three relations:

SUPPLY =

SUPPLIER_PART

JOIN SUPPLIER_PROJECT

JOIN PART_PROJECT

Step 3: Final 5NF Decomposition

SUPPLIER_PART(Supplier, Part)

SUPPLIER_PROJECT(Supplier, Project)

PART_PROJECT(Part, Project)

Advantages

- Removes join dependency.
- Reduces redundancy.
- Prevents insertion anomalies.
- Prevents deletion anomalies.
- Prevents update anomalies.

Therefore, the relations are decomposed into 5NF / **Project-Join Normal Form**.

Q9. Given a poorly designed database schema for a college, identify all anomalies (insertion, deletion, update) and normalize it to 3NF.

Answer:

Consider the poorly designed relation:

COLLEGE(StudentID, StudentName, CourseID, CourseName, FacultyID, FacultyName)

Example:

StudentID	StudentName	CourseID	CourseName	FacultyID	FacultyName
S01	Arun	C01	DBMS	F01	Kumar
S02	Priya	C01	DBMS	F01	Kumar
S03	Ravi	C02	OS	F02	Ravi

Functional Dependencies

StudentID -> StudentName

CourseID -> CourseName, FacultyID

FacultyID -> FacultyName

Therefore:

CourseID -> FacultyID -> FacultyName

This is a transitive dependency.

1. Insertion Anomaly

Suppose a new course is introduced but no student has enrolled.

The course information cannot be stored independently because the table expects student information.

Therefore:

Insertion Anomaly:

Cannot insert course information independently.

2. Deletion Anomaly

Suppose the only student enrolled in a particular course is deleted.

The course and faculty information may also be deleted.

Therefore:

Deletion Anomaly:

Deleting a student can unintentionally delete course/faculty information.

3. Update Anomaly

Suppose faculty F01 changes their name.

The faculty name appears in multiple rows.

All occurrences must be updated.

Therefore:

Update Anomaly:

The same information must be updated in multiple rows.

Normalize to 3NF

Table 1: STUDENT

STUDENT(StudentID, StudentName)

Primary key:

StudentID

Table 2: COURSE

COURSE(CourseID, CourseName, FacultyID)

Primary key:

CourseID

Table 3: FACULTY

FACULTY(FacultyID, FacultyName)

Primary key:

FacultyID

Table 4: ENROLLMENT

ENROLLMENT(StudentID, CourseID)

Primary key:

(StudentID, CourseID)

Final 3NF Schema

STUDENT(StudentID, StudentName)

COURSE(CourseID, CourseName, FacultyID)

FACULTY(FacultyID, FacultyName)

ENROLLMENT(StudentID, CourseID)

Result

The 3NF decomposition:

- Removes redundancy.
- Removes insertion anomaly.
- Removes deletion anomaly.
- Removes update anomaly.
- Removes transitive dependency.
- Improves data integrity.

Q10. Compute the closure of attribute sets for the given FDs and determine all candidate keys for the relation.

Answer:

Consider:

$R(A, B, C, D, E)$

Functional dependencies:

$A \rightarrow B$

$B \rightarrow C$

$CD \rightarrow E$

$E \rightarrow A$

Step 1: Calculate A^+

Start:

$A^+ = \{A\}$

Using:

$A \rightarrow B$

we get:

$A^+ = \{A, B\}$

Using:

$B \rightarrow C$

we get:

$A^+ = \{A, B, C\}$

We cannot use $CD \rightarrow E$ because D is missing.

Therefore:

$A^+ = \{A, B, C\}$

So A is not a candidate key.

Step 2: Calculate D^+

Start:

$D^+ = \{D\}$

No functional dependency can be applied.

Therefore:

$D^+ = \{D\}$

So D is not a candidate key.

Step 3: Calculate E^+

Start:

$E^+ = \{E\}$

Using:

$E \rightarrow A$

we get:

$E^+ = \{E, A\}$

Using:

$A \rightarrow B$

we get:

$E^+ = \{E, A, B\}$

Using:

$B \rightarrow C$

we get:

$E^+ = \{E, A, B, C\}$

D is still missing.

Therefore:

$E^+ = \{E, A, B, C\}$

So E is not a candidate key.

Step 4: Calculate (AD)+

Start:

$(AD)^+ = \{A, D\}$

Using:

$A \rightarrow B$

we get:

$(AD)^+ = \{A, B, D\}$

Using:

$B \rightarrow C$

we get:

$(AD)^+ = \{A, B, C, D\}$

Now C and D are present.

Using:

$CD \rightarrow E$

we get:

$(AD)^+ = \{A, B, C, D, E\}$

The closure contains all attributes.

Therefore:

AD is a superkey.

Neither A nor D alone is a superkey.

Therefore:

AD is a candidate key.

Final Table

Attribute Set	Closure	Candidate Key
A	{A, B, C}	No
D	{D}	No
E	{E, A, B, C}	No
AD	{A, B, C, D, E}	Yes

Final Answer

Candidate Key = AD